

# MIHIR UPASANI

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## Education

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### New York University, Tandon School of Engineering

New York, NY

MS, Computer Engineering | GPA: 3.4/4

Aug 2021 - May 2023

- **Coursework:** Machine Learning, Deep Learning, High Performance Machine Learning, Data Structures and Algorithms, Probability and Stochastic Processes

### Vishwakarma Institute of Technology

Pune, India

BTech, Computer Engineering | GPA: 7.67/10

Aug 2014 - May 2018

## Work Experience

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### DataNimbus

Jul 2023 - Present

ML Engineer

Plano, TX

- ML Experimentation Platform (Marketing Data Company): Building migration of legacy Java ML pipelines to Python on Databricks for ~30 data scientists. Three-stage pipeline: GLM feature selection -> H2O AutoML binary classification -> batch scoring. Training parallelized at 1,200 concurrent jobs in batches of 300. Built parity validation framework comparing feature importances, GLM coefficients, and model metrics (AUC, Gini, logloss) within 5% tolerance across 10k+ models. Converted orchestration to Databricks Asset Bundles with CI/CD. V1 piloting September 2026.
- Finhub UniTrack - Agentic Financial Intelligence Platform: Designed full-stack platform with React frontend, TypeScript backend, PostgreSQL, and MCP SDK. Multi-persona chat agent (Data Explorer, Financial Analyst, HR Assistant) driven entirely by system prompts. Agentic Double Diamond risk analysis: four-phase workflow running ~20 SQL aggregations, modeling pessimistic/baseline/optimistic scenarios, delivering structured risk cards with full query provenance. 5-9 risk insights per run.
- Demand Forecasting (Home Services Company): Built 150 per-business-unit models predicting hourly job/call volume over 14-day horizons across 50+ brands. Domain-specific feature engineering: weather lag, temperature-threshold triggers, seasonal indicators. 130/150 models at <10% daily MAPE. Powers live dispatch planning operations.
- Model Monitoring & Retraining Pipeline: Built monitoring layer comparing predictions vs. actuals daily across 200+ business units. Automated retrain triggers at 20% deviation threshold. Eliminated 6 hours of manual effort per reporting cycle. Zero failures since deployment.
- ML Model Suite - Home Services Company (Churn, LTV, Cold Call, Segmentation): Built models across 50+ brands serving 10k+ customers feeding a marketing AI platform. Churn: time-expanded cross-validation, 0.7 AUC under 0.25 max FPR. Segmentation: K-means with SHAP explainability. Cold Call Propensity: daily call lists per brand.
- GenAI Review Response System - Home Services Company: Automated Google review responses using Llama 3 with Llama Guard guardrails, 500+ responses/day across 50 brands. Human-in-the-loop queue for sub-4-star ratings. ~80% posted as-is. Reduced brand manager workload by 50%.
- Custom MCP Server: Multi-tenant server for AI agent database verification. Factory pattern supporting 6 adapters (MongoDB, PostgreSQL, MySQL, Azure SQL, BigQuery, SQLite). Read-only query validation, auto-schema discovery, SSH tunneling. Designed as a verifier agent in a multi-agent swarm.
- RDS to MongoDB Migration: Migrated demand forecasting app from Amazon RDS (\$5k/month) to MongoDB (\$1k/month). Full TypeScript/Node.js backend rewrite. 80% infra cost reduction (~\$48k/year savings).
- Solar Farm Predictive Maintenance (Solar O&M Company): Built inverter failure prediction models across 50 plants, 14-day horizon, 5 years of minute-by-minute sensor data. <0.3 FPR at 0.6 AUC.

### DataNimbus

May 2023 - Jul 2023

SWE Intern

New York, NY

- Dockerized React frontend and Python backend of Overread.io, containerizing ~50% of modules.
- Built Jenkins CI/CD pipelines replacing 10-step manual deployment, cutting time from ~10min to ~5min per module.
- Implemented non-destructive ECG document markup preserving source data integrity. Converted to full-time ML Engineer based on contributions.

### Persistent Systems

Jan 2021 - Jun 2021

Senior Software Engineer

Pune, India

- Solo-built v1 of Overread.io: patients submit single-lead ECGs from Fitbit, Apple Watch, Kardia; system extracts waveform data, heart rate, and cardiac metrics.
- Built computer vision pipeline using OpenCV to read seven-segment displays from HbA1c and blood pressure monitor photos.

- Designed microservices architecture processing 1,000s of ECGs/day with role-based access and multi-channel delivery (email, WhatsApp).

## Capiot Software

Jan 2020 - Jan 2021

*Associate Consultant*

*Pune, India*

- Built intelligent MDM system for a Hotel Booking Aggregator: room name deduplication using word2vec, Word Mover's Distance, cosine similarity, Levenshtein, and Double Metaphone. Ensemble model with human review loop. 70% accuracy, 25% reduction in manual MDM work.

## Capiot Software

Jun 2018 - Jan 2020

*Junior Consultant*

*Pune, India*

- Built address cleaning pipeline for an Indian Logistics Company processing 6M+ addresses/week via RabbitMQ, reducing returned packages by ~5%. N-gram hashes with Double Metaphone pronunciation-based dictionary.
- Promoted to Associate Consultant after 18 months — tied to successful delivery of projects for the logistics and hotel booking clients.

## Personal Projects

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### CIFAR-10 ResNet Hyperparameter Tuning | *PyTorch, CUDA*

- 95% test accuracy on CIFAR-10 with <5M parameters using modified ResNet-18 ([2,1,1,1] block config). Systematic search across 15+ experiments: 85% baseline -> 90% (LR scheduling) -> 94% (cyclic momentum) -> 95% (exponential decay at 100 epochs).
- [github.com/MihirUpasani/CIFAR10-ResNet18](https://github.com/MihirUpasani/CIFAR10-ResNet18)

### CIFAR-10 ResNet-50 with Test-Time Augmentation | *PyTorch, CUDA, Multi-GPU*

- Full ResNet-50 (Bottleneck [3,4,6,3]) with SGD + Nesterov momentum, ReduceLROnPlateau. Multi-GPU support. Achieved 92%+ TTA accuracy.
- [github.com/MihirUpasani/CIFAR10-TTA92-ResNet](https://github.com/MihirUpasani/CIFAR10-TTA92-ResNet)

### Learning Rate vs Batch Size Experiments | *Python, PyTorch, Jupyter*

- Reproduction of Smith et al. 'Don't Decay the Learning Rate, Increase the Batch Size' (arXiv:1711.00489). Empirically validated batch size scaling as substitute for LR decay. Implemented cyclical LRs, analyzed noise ratio (LR / batch size) as governing variable.
- [github.com/MihirUpasani/LearningRatevsBatchSize](https://github.com/MihirUpasani/LearningRatevsBatchSize)

## Technical Skills

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**Languages:** Python, TypeScript, JavaScript, SQL

**ML/AI:** PyTorch, H2O AutoML, Databricks AutoML, SHAP, K-means, word2vec, LLMs (Llama 3), Llama Guard, GLM, ensemble methods

**Data & Platforms:** Databricks, MongoDB, PostgreSQL, MySQL, SQLite, BigQuery, Pandas, RabbitMQ

**Cloud & Infra:** AWS (EC2, RDS, S3), Docker, Kubernetes, CI/CD, Jenkins, Databricks Asset Bundles

**Frameworks:** React, Node.js, asyncio, gunicorn, Pydantic, MCP SDK

**Certifications:** Databricks Certified Data Engineer Associate (2023), Databricks Certified Data Engineer Professional (2023), Databricks Certified Machine Learning Associate (2023)